

Lab 8 : Implementation and Performance Evaluation of Categorical Naive Bayes Classifier

Develop a complete Python program to build a classification pipeline using the standard **Play Tennis dataset**. Your implementation must complete the following sequential tasks:

https://docs.google.com/spreadsheets/d/1mnoUgK8YxInzoDQ7P7iBM1HeZ8tcnUSvMU_H1OWndFA/edit?gid=0#gid=0

1. **Data Preprocessing:**
 - Load the dataset and separate the input features (**Outlook**, **Temperature**, **Humidity**, **Wind**) from the target variable (**Play**).
 - Convert all categorical feature values and target labels into numerical representations using an appropriate encoding technique.
2. **Dataset Partitioning:**
 - Divide the dataset into training and testing subsets using an 80:20 train-test split ratio.
3. **Naive Bayes Model Training & Evaluation:**
 - Train a **Categorical Naive Bayes (CategoricalNB)** model on the training data.
 - Predict the class labels for the test dataset.
 - Calculate and display the overall **Model Accuracy**.
 - Display the **Confusion Matrix** and **Classification Report** (Precision, Recall, F1-Score).
4. **Single-Sample Inference:**
 - Predict whether a person will play tennis under the following specific weather conditions:
 - **Outlook:** Sunny
 - **Temperature:** Cool
 - **Humidity:** High
 - **Wind:** Strong
 - Display both the **predicted class label** and the **corresponding class probabilities**
5. **Model Comparison:**
 - Train a **Decision Tree Classifier** and a **Logistic Regression Classifier**, and an **SVM** on the same training data.
 - Compare all three models in terms of test accuracy, single-sample query prediction, and output class probabilities.

DELIVERABLES

- **Executable Python Script (.py or .ipynb):** Clean, well-commented source code fulfilling all tasks.
- **Output Log:** Console output showing the Naive Bayes evaluation metrics, custom query prediction with probabilities, and the model comparison table.

- **Analysis Report:** A brief explanation (3–4 sentences) comparing why the models produced different predictions and probability scores for the given test instance.